

# ASAD IQBAL

Software Engineering Graduate | Full-Stack & Backend Developer | AI & Computer Vision

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## PROFESSIONAL SUMMARY

Software Engineering graduate with hands-on experience in **full-stack web development, backend engineering, REST APIs, Artificial Intelligence, Machine Learning, and Computer Vision**. Experienced in building and deploying applications using **React.js, Node.js, Express.js, FastAPI, ASP.NET Core, Firebase, PostgreSQL, and MongoDB**. Completed an intensive **8-week AI internship** covering machine learning, deep learning, computer vision, OCR, FastAPI, and AI application deployment. Experienced as a **Backend Lead** on a deployed full-stack business platform, developing APIs, authentication, security, database integration, and server-side business logic. Interested in opportunities in **software development, backend engineering, full-stack development, and AI**.

## TECHNICAL SKILLS

**Programming Languages:** C, C++, Java, C#, Python, JavaScript

**Frontend Development:** HTML5, CSS3, JavaScript, React.js, Vite, Responsive Web Design

**Backend Development:** Node.js, Express.js, FastAPI, ASP.NET Core, Razor Pages, REST APIs

**AI & Machine Learning:** Machine Learning, Neural Networks, Deep Learning, Computer Vision, YOLO, PyTorch, Ultralytics, EasyOCR, OpenCV, Scikit-learn

**Data & Visualization:** NumPy, Pandas, Matplotlib

**Databases:** MongoDB, PostgreSQL, MySQL, Firebase Firestore

**Authentication & Security:** Firebase Authentication, Firebase Admin SDK, ID Token Verification, CORS, Rate Limiting, Server-side Validation

**Tools & Deployment:** Git, GitHub, Postman, Vercel, Hugging Face Spaces, Swagger/OpenAPI

## PROJECTS

### ANPR Vision AI — AI-Powered License Plate Recognition

React.js | Vite | FastAPI | Python | YOLO | EasyOCR | OpenCV | PyTorch

- Built and deployed a full-stack AI application for detecting vehicle license plates and recognizing plate numbers from uploaded images.
- Implemented **YOLO-based license plate detection** and **EasyOCR** for extracting characters from detected plates.
- Developed a **FastAPI REST API** for image upload, validation, AI processing, bounding-box coordinates, JSON responses, and error handling.
- Integrated the AI backend with a **React/Vite frontend** for displaying detection and OCR results.
- Implemented responsive UI, loading states, error handling, and end-to-end API integration.
- Deployed the frontend on **Vercel** and FastAPI backend on **Hugging Face Spaces**.
- Tested API functionality using **Postman** and documented the project workflow on GitHub.

Live: [anpr-vision-ai.vercel.app/](https://anpr-vision-ai.vercel.app/)

### MIFRA Enterprises — Full-Stack Business Platform

FastAPI | Python | React.js | Firebase | Firestore | Firebase Admin SDK | REST APIs | Vercel

- **Role: Backend Lead**
- Led backend development of a **deployed full-stack business platform** using FastAPI.
- Designed and implemented backend architecture, server-side business logic, API routing, validation, security, and integration.
- Developed REST APIs for **products, categories, services, product requests, service requests, contact messages, site settings, and admin operations**.
- Integrated **Firebase Admin SDK and Firestore** for application data management and backend services.
- Implemented **Firebase ID token verification** and protected admin APIs for authenticated operations.
- Developed CRUD APIs for products and services, including **stock management and administrative operations**.
- Implemented customer request and contact APIs with server-side validation and structured error handling.
- Configured **CORS, rate limiting, authentication verification, and backend security**.
- Integrated backend APIs with the public React website and separate React admin panel.
- Tested APIs using **Postman** and provided API documentation through **Swagger/OpenAPI**.
- Provided technical guidance to team members on backend, API, authentication, and security-related issues.

Live: [mifra-enterprises-website.vercel.app/](https://mifra-enterprises-website.vercel.app/) | API Docs: [mifra-enterprises-website.vercel.app/docs](https://mifra-enterprises-website.vercel.app/docs) | Admin: [mifra-enterprises-admin.vercel.app/](https://mifra-enterprises-admin.vercel.app/)

## BIIT Sports Tournament & Live Scoring App

ASP.NET Core | Razor Pages | JavaScript | Spring Boot | PostgreSQL | REST APIs | WebSockets

- Developed a multi-sport tournament management and live scoring platform for **BIIT Sports Society**.
- Designed workflows for **team registration, tournament management, fixtures, player statistics, points tables, and live scoring**.
- Built responsive web interfaces using **ASP.NET Core Razor Pages**.
- Integrated the frontend with a **Spring Boot REST API** and PostgreSQL backend.
- Implemented **WebSocket-based real-time match updates** for live scoring.
- Designed support for multiple sports including **Cricket, Futsal, Badminton, Table Tennis, Volleyball, Chess, Ludo, and Tug-of-War**.

## Expense Tracker — MERN Full-Stack Application

MongoDB | Express.js | React.js | Node.js | JavaScript

- Developed a full-stack expense tracking application using the **MERN stack**.
- Implemented user authentication and CRUD functionality for income and expense records.
- Developed backend APIs for transaction management and dashboard data.
- Built a React frontend with reusable components and REST API integration.
- Implemented financial data visualization using charts.
- Used MongoDB for persistent data storage and Express/Node.js for backend services.

## EXPERIENCE

### AI Internship — 8 Weeks

Artificial Intelligence | Machine Learning | Computer Vision | Backend Development

- Completed an intensive 8-week AI internship covering **Python, data analysis, machine learning, neural networks, deep learning, computer vision, OCR, API development, and deployment**.
- Worked with **NumPy, Pandas, Matplotlib, and Scikit-learn** for data processing, analysis, visualization, and machine learning.
- Worked with **PyTorch, YOLO, Ultralytics, OpenCV, and EasyOCR** for AI and computer vision applications.
- Developed AI functionality using **FastAPI REST APIs**.
- Integrated AI backends with web applications and deployed applications online.
- Used **GitHub and Postman** for project development, documentation, and API testing.

## EDUCATION

### Bachelor of Science in Software Engineering

PMAS-Arid Agriculture University, Rawalpindi, Pakistan

Graduated — 2026

## CORE AREAS

Full-Stack Web Development • Backend Development • REST API Development • Artificial Intelligence • Machine Learning • Computer Vision • React.js Development • ASP.NET Core Development • FastAPI Development • Software Engineering

## LANGUAGES

English | Urdu